

19/7/26.

Data Structure

Home Work Note

1) Sum of 2 Matrix

```
#include <stdio.h>
```

```
{ int a[20][20], b[20][20];
```

```
int d[20][20], i, j, c=0, r=0;
```

```
printf("Enter no. of rows and columns = ");
```

```
scanf("%d %d", &r, &c);
```

```
printf("Enter Matrix 1 = ");
```

```
for(i=0; i<r; i++)
```

```
{ for(j=0; j<c; j++)
```

```
{ scanf("%d", &a[i][j]);
```

```
} printf("Enter Matrix 2 = ");
```

```
for(i=0; i<r; i++)
```

```
{ for(j=0; j<c; j++)
```

```
{ scanf("%d", &b[i][j]);
```

```
} for(i=0; i<r; i++)
```

```
{ for(j=0; j<c; j++)
```

```
{ d[i][j] = a[i][j] + b[i][j];
```

```
} }
```

```
printf("\n Sum = ");
```

```
for(i=0; i<r; i++)
```

```
{ for(j=0; j<c; j++)
```

```
{ printf("\t%d", d[i][j]);
```

```
} printf("\n");
```

```
}
```

```
return 0;
```

```
}
```

Enter no. of rows and columns =
2 2

Enter matrix 1 =

1	2
3	4

Enter matrix 2 =

1	2
3	4

Sum =

4

8

- 2) Write a program in C to read n number of value in an array & reverse display them in reverse order.

```
#include <stdio.h>
```

```
int main()
```

```
{ int a[20], i;  
  int n;
```

```
  printf("Enter n = ");
```

```
  scanf("%d", &n);
```

```
  printf("Enter elements = ");
```

```
  for(i=0; i<n; i++)
```

```
  { scanf("%d", &a[i]);
```

```
  }
```

```
  printf("\n reverse order = ");
```

```
  for(i=n-1; i>=0; i--)
```

```
  { printf("%d", a[i]);
```

```
  }
```

```
  return 0;
```

```
}
```

O/p Enter n = 4

Enter elements = 1

2 3 4

reverse order =

4 3 2 1

- 3) Count the total number of duplicate elements in an array.

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
  int a[20], b[10], i;
```

```
  int n, count = 1, j;
```

```
  printf("Enter n = ");
```

```
  scanf("%d", &n);
```

```
  printf("Enter elements = ");
```

```
  for(i=0; i<n; i++)
```

```
  { scanf("%d", &a[i]);
```

```
  }
```

```
  printf("\n duplicate elements = ");
```



```
for (i = 0; i < n; i++)
```

```
{ count = 1;
```

```
  for (j = i + 1; j < n; j++)
```

```
  { if (a[i] == a[j])
```

```
    count++;
```

```
  }
```

```
  b[i] = count;
```

```
}
```

```
for (i = 0; i < n; i++)
```

```
{ if (b[i] != 1)
```

```
  { printf("\n %d repeats %d times", a[i], b[i]);
```

```
  }
```

```
return 0;
```

```
}
```

O/P Enter n = 4

Elements = 1 1 2 2

duplicate elements = 1

1 repeats 2 times

2 repeats 2 times

main()

```
{
```

```
  int a[20], i, n, count = 0, j;
```

```
  printf("Enter n = ");
```

```
  scanf("%d", &n);
```

```
  printf("Enter elements = ");
```

```
  for (i = 0; i < n; i++)
```

```
  { scanf("%d", &a[i]);
```

```
  }
```

```
  for (i = 0; i < n; i++)
```

```
  { for (j = i + 1; j < n; j++)
```

```
    { if (a[i] == a[j])
```



```

    count++;
}
}
printf("No. of duplicate elements = %d", count);
return 0;
}

```

O/p Enter n = 5

Enter elements = 1 2 3 2 1

No of duplicate elements = 2

4) Print unique elements

```

#include <stdio.h>
int main()
{
    int a[20], i, j;
    int n, count;
    printf("Enter n = ");
    scanf("%d", &n);
    printf("Enter elements = ");
    for(i=0; i<n; i++)
    {
        scanf("%d", &a[i]);
    }

```

O/p

Enter n = 4

Enter elements 1 2
2 3

```

    for(i=0; i<n; i++)
    {
        count = 0;
        for(j=0; j<n; j++)
        {
            if (i != j && a[i] == a[j])
                count++;
        }

```

```

    if (count == 0)
    {
        printf("%d", a[i]);
    }
}
return 0;
}

```


5.

Insert in Sorted array.

#include <stdio.h>

int main()

{ int a[20], i, n;

printf("Enter n = ");

scanf("%d", &n);

printf("Enter sorted array = ");

for(i=0; i<n; i++)

{ scanf("%d", &a[i]);

}

int item;

printf("\n Enter number to be inserted = ");

scanf("%d", &item);

i = n-1;

while(i >= 0 && a[i] > item)

{ a[i+1] = a[i];

i--;

}

a[i+1] = item;

n++;

printf("\n After insertion = ");

for(i=0; i<n; i++)

{ printf("%d", a[i]);

}

return 0;

}

Q/P Enter n = 5

Enter sorted array = 2 3 4 5 7

Enter number to be inserted = 6

After insertion = 2 3 4 5 6 7

7) Matrix Multiplication

```
#include <stdio.h>
```

```
int main()
```

```
{
    int a[10][10], b[10][10], d[10][10];
```

```
    int r1, c1, r2, c2;
```

```
    int i, j, k;
```

```
    printf("Enter rows & columns of matrix 1 = ");
```

```
    scanf("%d %d", &r1, &c1);
```

```
    printf("Enter rows & columns of matrix 2 = ");
```

```
    scanf("%d %d", &r2, &c2);
```

```
    if (c1 != r2)
```

```
    {
        printf("\n Multiplication Not possible");
    }
```

```
    else
```

```
    {
        printf("Enter Matrix 1 = ");
```

```
        for (i = 0; i < r1; i++)
```

```
        {
            for (j = 0; j < c1; j++)
```

```
            {
                scanf("%d", &a[i][j]);
            }
        }
```

```
        printf("\n Enter Matrix 2 = ");
```

```
        for (i = 0; i < r2; i++)
```

```
        {
            for (j = 0; j < c2; j++)
```

```
            {
                scanf("%d", &b[i][j]);
            }
        }
```

```
        printf("\n Matrix Multiplication = \n");
```

```
        for (i = 0; i < r1; i++)
```

```
        {
            for (j = 0; j < c2; j++)
```

```
            {
                d[i][j] = 0;
```

```
                for (k = 0; k < r2; k++)
```

```
                {
                    d[i][j] += a[i][k] * b[k][j];
                }
            }
        }
```



```

    }
    }
    }
    for(i=0; i<r1; i++)
    {
        for(j=0; j<c2; j++)
        {
            printf(" %d", d[i][j]);
        }
        printf("\n");
    }
    // else
    return 0;
}

```

O/P Enter rows & columns of matrix 1 = 2, 3
 Enter rows & columns of matrix 2 = 3, 2

Enter Matrix 1 =

1	2	3
4	5	6

Enter Matrix 2 =

1	2
3	4
5	6

Matrix Multiplication =

22	28
49	64

8) Matrix Transpose :

```
#include <stdio.h>
```

```
int main()
```

```
{
    int i, r, c, j;
```

```
    int a[20][20];
```

```
    printf("Enter rows & cols = ");
```

```
    scanf("%d %d", &r, &c);
```

```
    printf("Enter Matrix = ");
```

```
    for(i=0; i<r; i++)
```

```
    {
        for(j=0; j<c; j++)
```

```
        {
            scanf("%d", &a[i][j]);
```

```
        }
    }
```



```

2
printf("\n Transpose Matrix = \n");
for(i=0; i<c; i++)
{
    for(j=0; j<r; j++)
    {
        printf("%d", a[j][i]);
    }
    printf("\n");
}
return 0;
}

```

O/P Enter rows & cols = 2 , 3

Enter Matrix =

1	2	3
4	5	6

Transpose Matrix =

1	4
2	5
3	6

9) Right Diagonal Sum:

```
#include <stdio.h>
```

```
int main()
```

```
{
    int i, n, j, sum = 0;
```

```
    int a[20][20];
```

```
    printf("Enter order = ");
```

```
    scanf("%d", &n);
```

```
    printf("\n Enter Matrix = ");
```

```
    for(i=0; i<n; i++)
```

```
    {
        for(j=0; j<n; j++)
```

```
        {
            scanf("%d", &a[i][j]);
        }
    }
```

```
    printf("\n Right Diagonal sum = ");
```

```
    for(i=0; i<n; i++)
```

```
    {
        sum += a[i][n-1-i];
    }
```

```
    printf("%d", sum);
```

```
    return 0;
```

```
}
```


O/P

Enter order = 3

Enter Matrix =

1	2	3
4	5	6
7	8	9

Right diagonal sum = 15

10)

Check for Identity Matrix

#include <stdio.h>

int main()

{ int i, j, n;

int a[20][20];

int identity = 1;

printf("Enter order = ");

scanf("%d", &n);

printf("Enter matrix = \n");

for(i=0; i<n; i++)

{ for(j=0; j<n; j++)

{ scanf("%d", &a[i][j]);

}

for(i=0; i<n; i++)

{ for(j=0; j<n; j++)

{ if(i==j)

{ if(a[i][j] != 1)

{ identity = 0;

}

else {

if(a[i][j] != 0)

{ identity = 0;

}

}

for(identity == 1)

{ printf("\n It is an Identity Matrix ");

else printf("\n It is not an Identity Matrix ");

return 0;

}

O/P Enter order = 3

Enter Matrix =

1	0	0
0	1	0
0	0	1

It is an Identity Matrix

To find no. of vowels, consonants, digits and white space in a string.

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main()
```

```
{ char a[100];
```

```
int vowel = 0, conso = 0, space = 0, digit = 0;
```

```
printf("Enter a string = ");
```

```
fgets(a, sizeof(a), stdin);
```

```
int len = strlen(a);
```

```
for(i = 0; i < len; i++)
```

```
{  
    if((a[i] >= 'A' && a[i] <= 'Z') ||  
        (a[i] >= 'a' && a[i] <= 'z'))
```

```
{  
    if(a[i] == 'a' || a[i] == 'e' || a[i] == 'i' ||
```

```
    a[i] == 'o' || a[i] == 'u' || a[i] == 'A' || a[i] == 'E' ||
```

```
    a[i] == 'I' || a[i] == 'O' || a[i] == 'U')
```

```
{  
    vowel++;
```

```
}  
else {  
    conso++;
```

```
}  
else if(a[i] >= '0' && a[i] <= '9')  
{  
    digit++;
```

```
}  
else if(a[i] == ' ')
```

```
{  
    space++;
```



```

printf("\n Vowel = %d\n Consonant = %d\n", vowel,
       Digits = %d\n white space = %d\n", vowel,
       conso, digit, space);
O/P Enter a string = asmi 14 years 56 high

```

```

Vowel = 4
Consonant = 7
Digits = 4
white space = 3

```

13) Count words in a string.

```

#include <stdio.h>
#include <string.h>
int main()
{
    char a[100];
    int space = 0;
    printf("Enter a string = ");
    fgets(a, sizeof(a), stdin);
    for(int i = 0; a[i] != '\0'; i++)
    {
        if(a[i] == ' ' && a[i+1] != ' ')
        {
            space++;
        }
    }
    printf("\n No. of words = %d", space + 1);
    return 0;
}

```

14) Sort string array (bubble sort)

```

#include <stdio.h>
int main()
{
    char a[5][20];
    char temp[20];
    int i, j;
    printf("Enter a string array = ");
}

```



```

for(i=0; i<5; i++)
    scanf("%d", a[i]);
}
for(i=0; i<5; i++)
    for(j=i+1; j<5; j++)
        if(strcmp(a[i], a[j])>0)
            strcpy(temp, a[i]);
            strcpy(a[i], a[j]);
            strcpy(a[j], temp);
    }
}
printf("\n sorted array = ")
for(i=0; i<5; i++)
    printf("\n %s", a[i]);
return 0;
}

```

o/p Enter a string array = asmi
 Ram
 Raju
 Kalika
 Moni

Sorted array = asmi
 Kalika
 Raju
 Ram
 Moni

15)

Toggle case of a sentence (Lower to upper & vice versa)

```

#include <stdio.h>
#include <string.h>

```

```

int main()

```

```

{
    void up-to-low(char a[]);
    void low-to-up(char a[]);

```

```

char a[20];

```

```

int n, i;

```

```

printf("Enter a string = ");

```



```
fgets(a, sizeof(a), stdin);  
printf("\n Enter 1 for UpperCase to Lower\n  
Enter 2 for lowercase to Upper\n");  
scanf("%d", &n);
```

```
switch(n)
```

```
{  
    case 1:
```

```
        up-to-low(a);
```

```
        break;
```

```
    case 2:
```

```
        low-to-up(a);
```

```
        break;
```

```
}
```

```
printf("\n After toggle ");
```

```
printf("\n %s", a);
```

```
return 0;
```

```
}
```

```
void up-to-low(char a[])
```

```
{  
    int i;
```

```
    for(i=0; a[i]!='\0'; i++)
```

```
    {  
        if(a[i] >= 'A' && a[i] <= 'Z')
```

```
        {  
            a[i] = a[i] + 32;
```

```
        }  
    }
```

```
}
```

```
void low-to-up(char a[])
```

```
{  
    int i;
```

```
    for(i=0; a[i]!='\0'; i++)
```

```
    {  
        if(a[i] >= 'a' && a[i] <= 'z')
```

```
        {  
            a[i] = a[i] - 32;
```

```
        }  
    }
```

```
}
```


O/P Enter a string = asmithe

Enter 1 for UpperCase to Lower

Enter 2 for Lowercase to Upper

2
After toggle

ASMITHA

12) Frequency of character in array

```
#include <stdio.h>
```

```
int main()
```

```
{  
    char a[20];
```

```
    int b[20], count = 0;
```

```
    int i, j;
```

```
    printf("Enter a string = ");
```

```
    scanf("%s", a);
```

```
    for(i = 0; a[i] != '\0'; i++)
```

```
    {  
        count = 1;
```

```
        if (a[i] != '\0')
```

```
        {  
            for(j = i + 1; a[j] != '\0'; j++)
```

```
            {  
                if (a[i] == a[j])
```

```
                {  
                    count += 1;
```

```
                    a[j] = '\0';
```

```
                }  
            }
```

```
            b[i] = count;
```

```
        }
```

```
    for(i = 0; a[i] != '\0'; i++)
```

```
    {  
        if (a[i] != '\0')
```

```
        {  
            printf("%c repeats %d times", a[i], b[i]);
```

```
        }
```

```
    return 0;
```

```
}
```

O/P Enter a string = array

a repeats 2 times

r repeats 2 times

y repeats 1 times